

Day 12: Regressions

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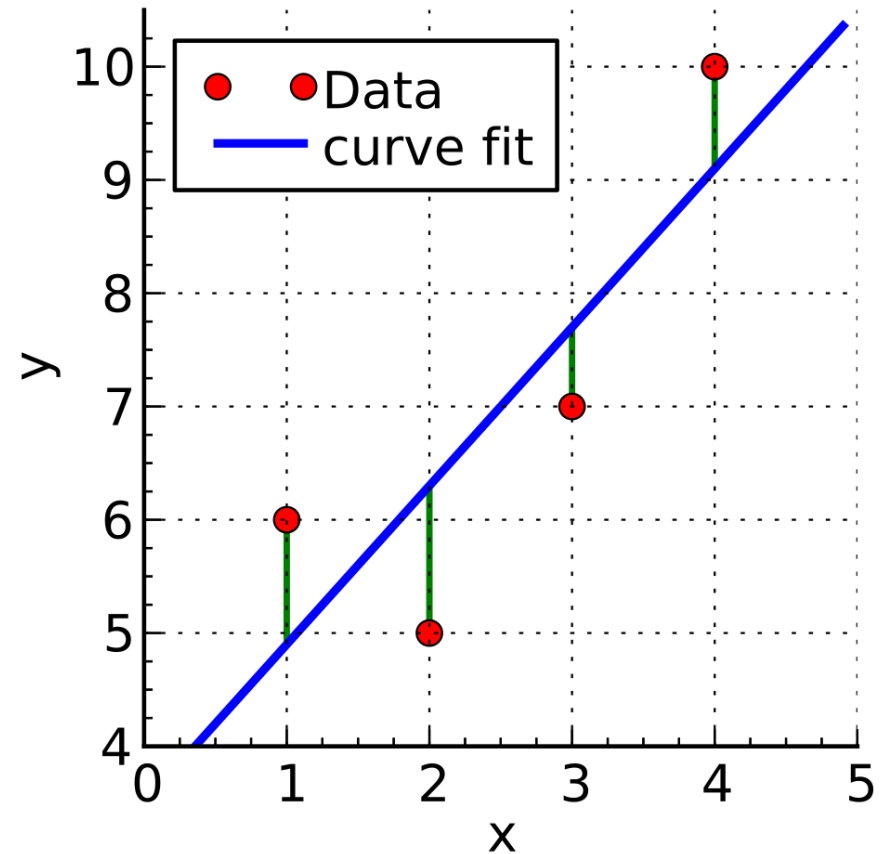
Linear regression

- Originally discovered in statistics field.
- The idea is to build a parametric, linear function that follow the multi-dimension data.

Ordinary Least Square

$$y = X\beta + \boxed{\varepsilon}$$

$$\beta = (X'^T X')^{-1} X'^T y$$



Polynomial regression

$$y = X'\beta + \varepsilon$$

$$X' = \begin{bmatrix} x_1^0 & \cdots & x_1^m \\ \vdots & \ddots & \vdots \\ x_n^0 & \cdots & x_n^m \end{bmatrix}$$

Polynomial regression

$$y = X'\beta + \varepsilon$$

$$X' = \begin{bmatrix} x_1^0 & \cdots & x_1^m \\ \vdots & \ddots & \vdots \\ x_n^0 & \cdots & x_n^m \end{bmatrix}$$


$$\beta = (X'^T X')^{-1} X'^T y$$

Degree
 $m \leq n$

Demo

x	y
1.5	-0.96
2	-2.71
3.5	-3.13
5	2.26
6.5	3.45
7	2.06

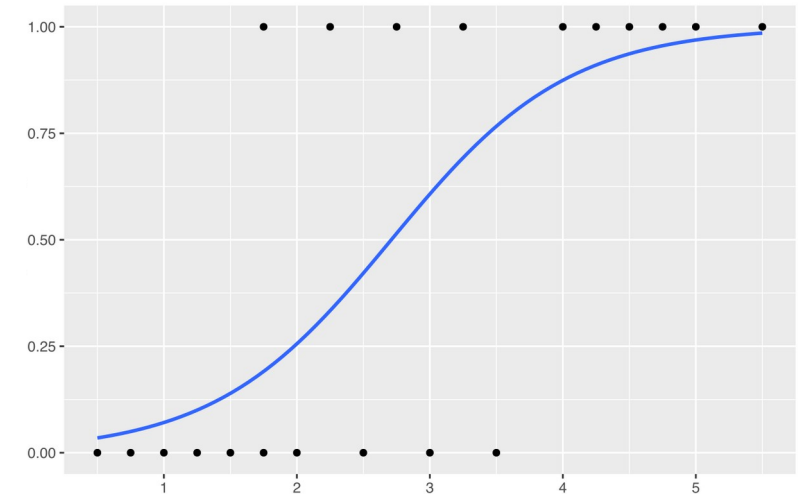
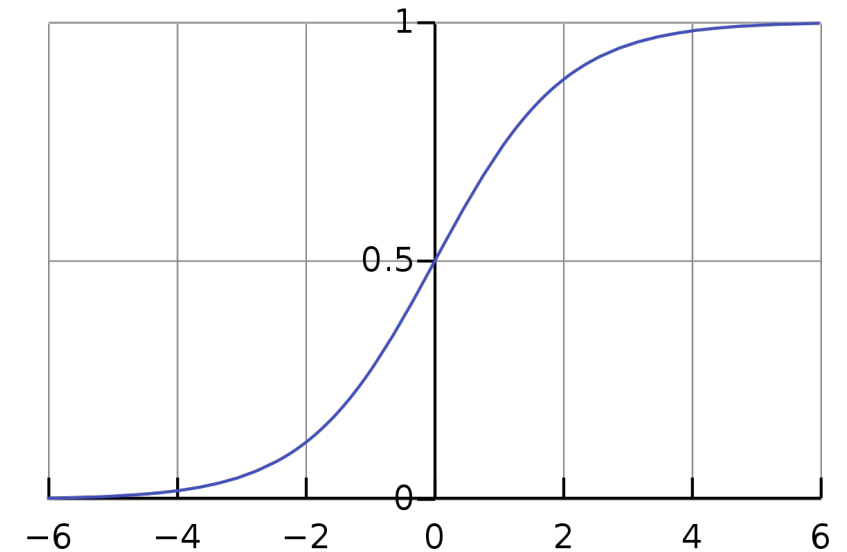
Logistics regression

- Make use of logistic function.

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}}$$

The actual model:

$$f(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$



$$l_{x_0} = \begin{cases} -\ln(f(x_0)) & \text{if } y_0 = 1 \\ -\ln(1 - f(x_0)) & \text{if } y_0 = 0 \end{cases}$$

$$l_{x_0} = -\ln(f(x_0)) - \ln(1 - f(x_0))$$

$$l_{x_0} = \begin{cases} -\ln(f(x_0)) & \text{if } y_0 = 1 \\ -\ln(1 - f(x_0)) & \text{if } y_0 = 0 \end{cases}$$

↓ $l_{x_0} = -y_0 \ln(f(x_0)) - (1 - y_0) \ln(1 - f(x_0))$

$$l_{x_0} = \begin{cases} -\ln(f(x_0)) & \text{if } y_0 = 1 \\ -\ln(1 - f(x_0)) & \text{if } y_0 = 0 \end{cases}$$

↓ $l_{x_0} = -y_0 \ln(f(x_0)) - (1 - y_0) \ln(1 - f(x_0))$

↑ $l = \sum_{i=1}^M y_i \ln(f(x_i)) + (1 - y_i) \ln(1 - f(x_i))$

Facts: Parametric regressions

- Parametric
 - Make assumption using math functions
 - Rigid
 - Make use of mathematics approaches
 - Overfit avoidance

II. Advanced approaches

- Kernel regression
- Decision tree regression
- Gradient boosting

Kernel regression

$$f(x) = \frac{ax}{b}$$

$$f \propto ax$$

Kernel regression

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f \propto e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f \propto \frac{1}{2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(Symmetric)

x

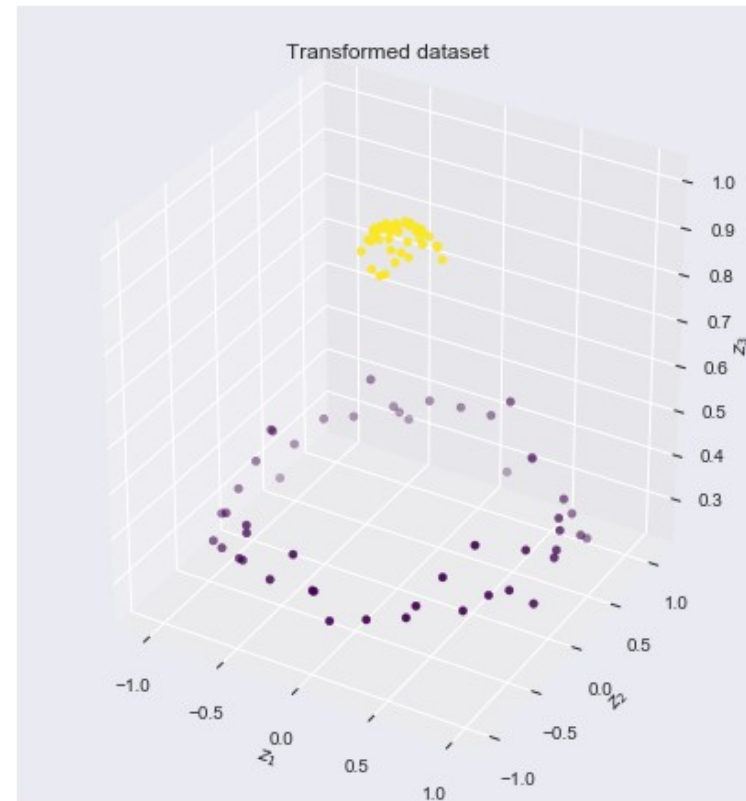
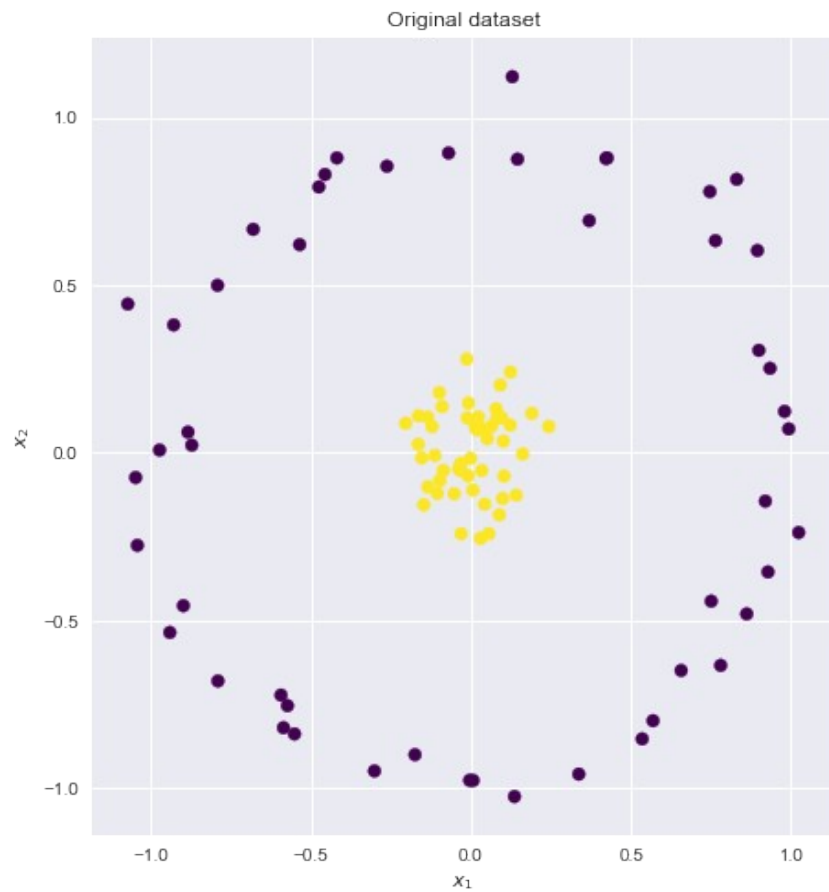


x'



Gaussian Radial Basis Function (RBF) centered at 0,0

$$\phi(x_1, x_2) = (z_1, z_2, z_3) = (x_1, x_2, e^{-[x_1^2 + x_2^2]})$$

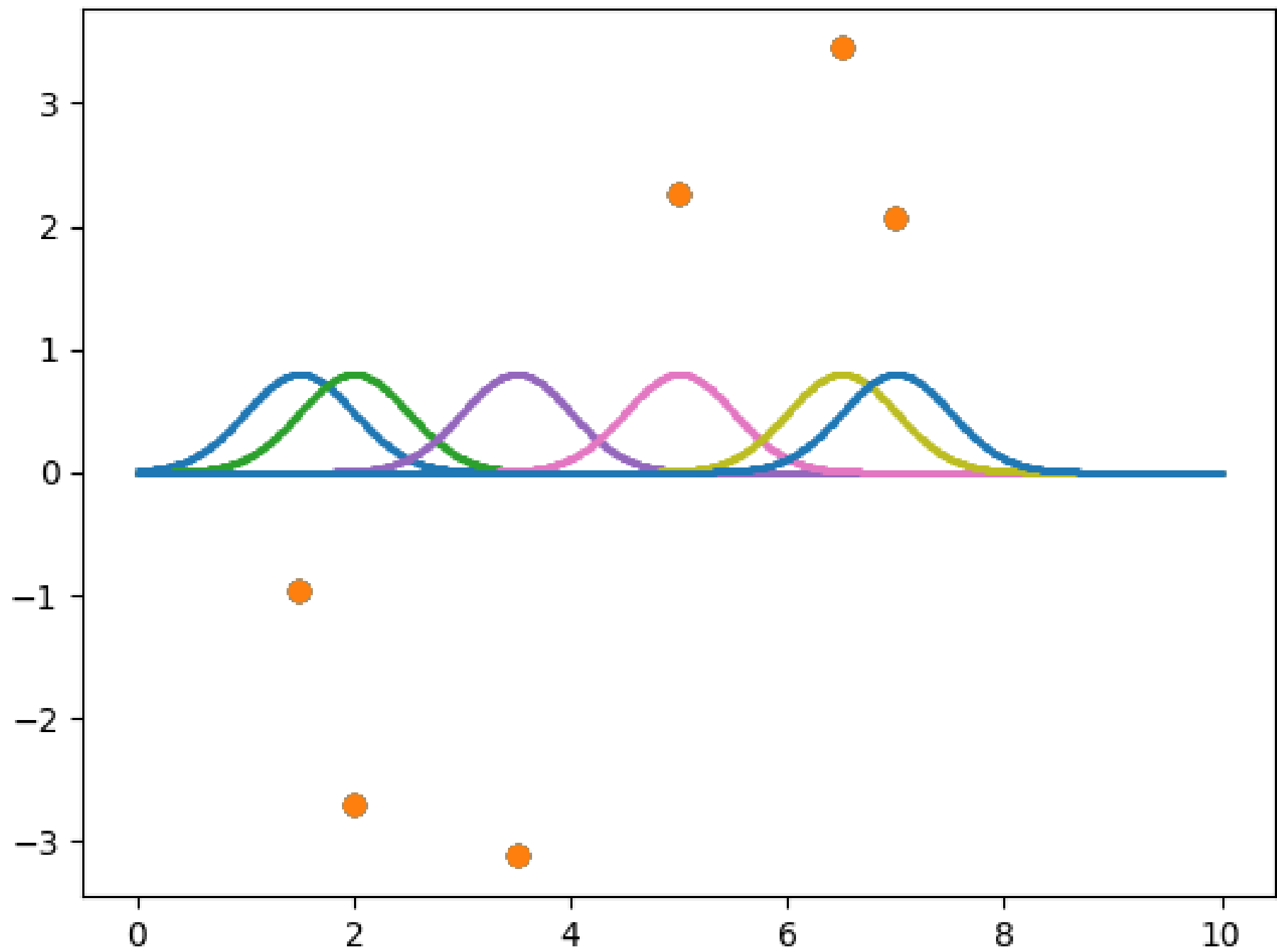


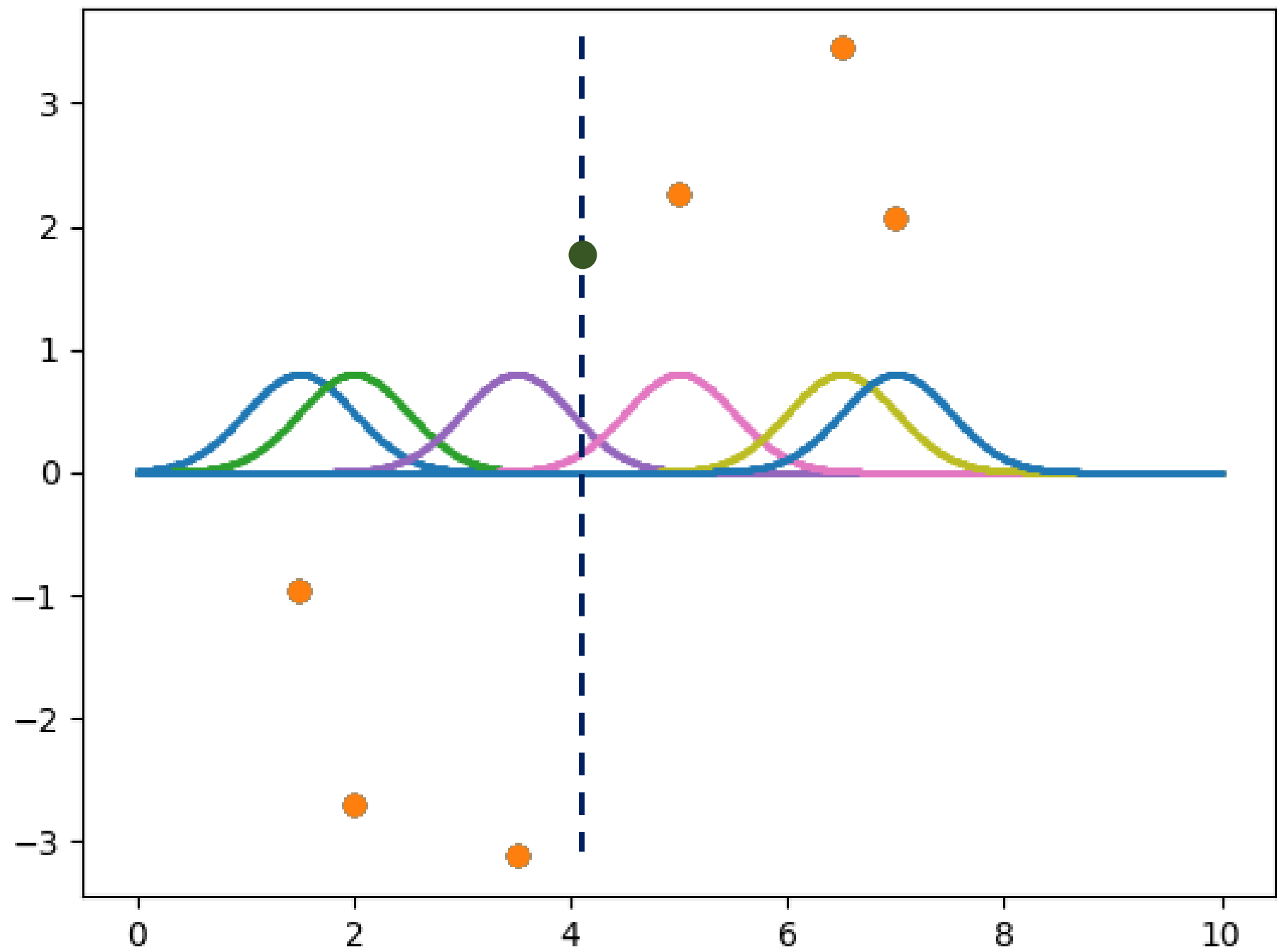
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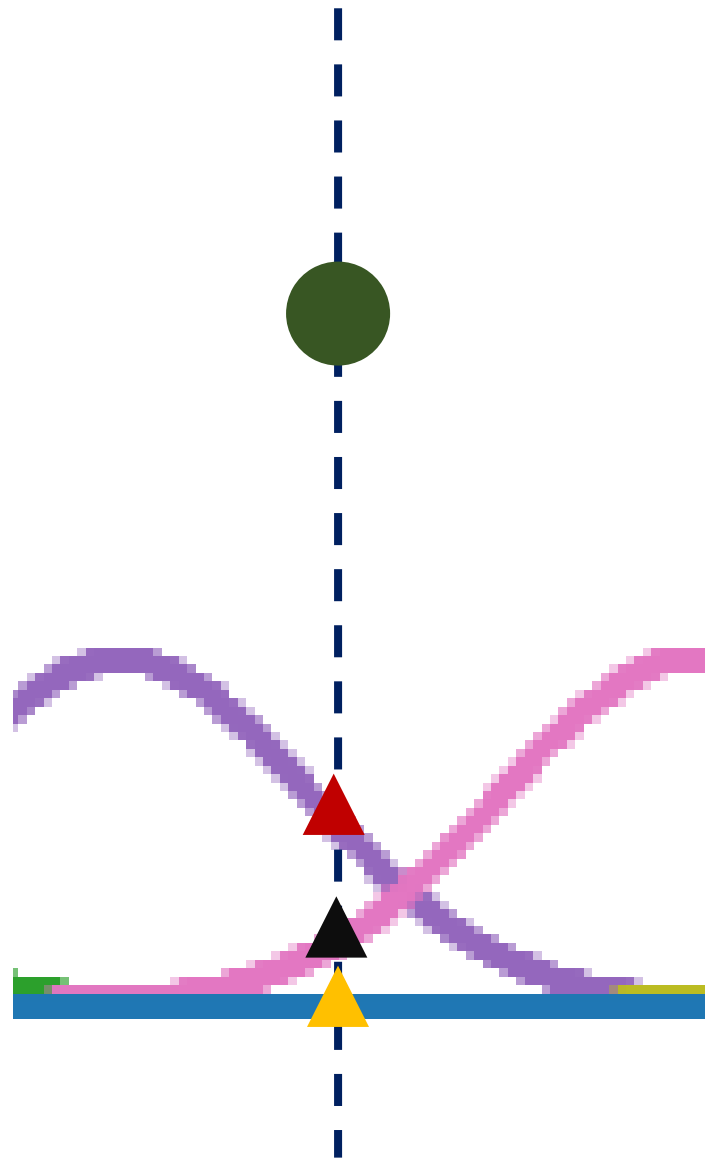
Gaussian kernel

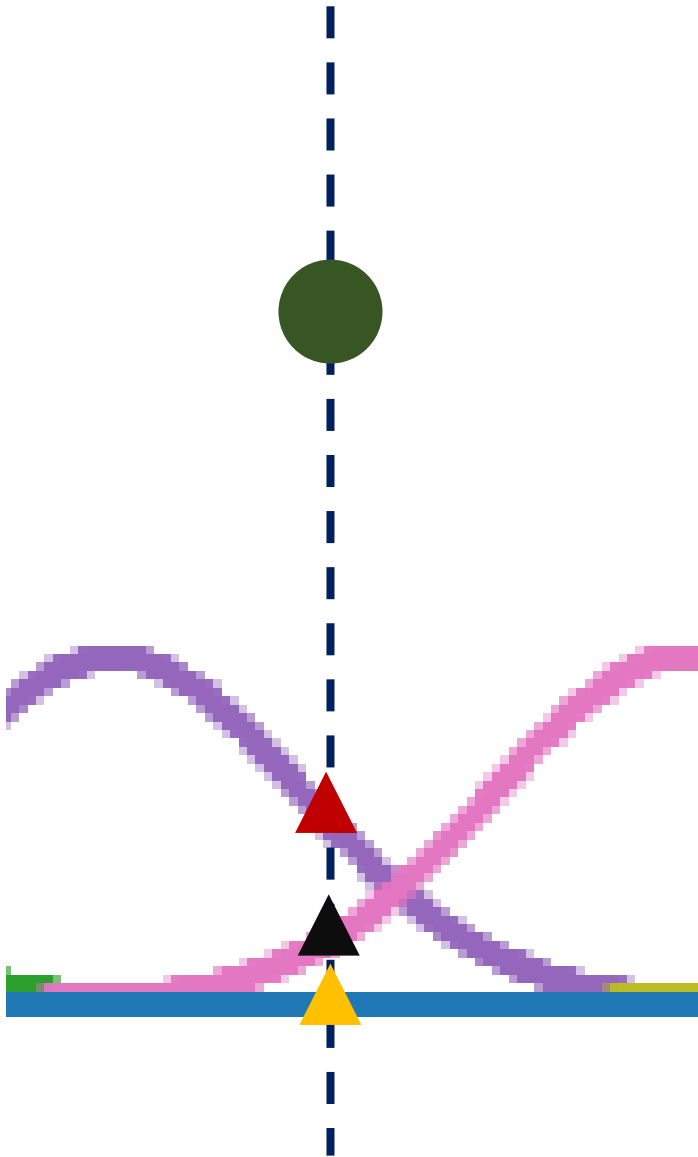
$$k_i(x|h, x_i) = \frac{1}{h\sqrt{2\pi}} e^{-0.5\left(\frac{x-x_i}{h}\right)^2}$$

x	y	h
1.5	-0.96	2
2	-2.71	2
3.5	-3.13	2
5	2.26	2
6.5	3.45	2
7	2.06	2

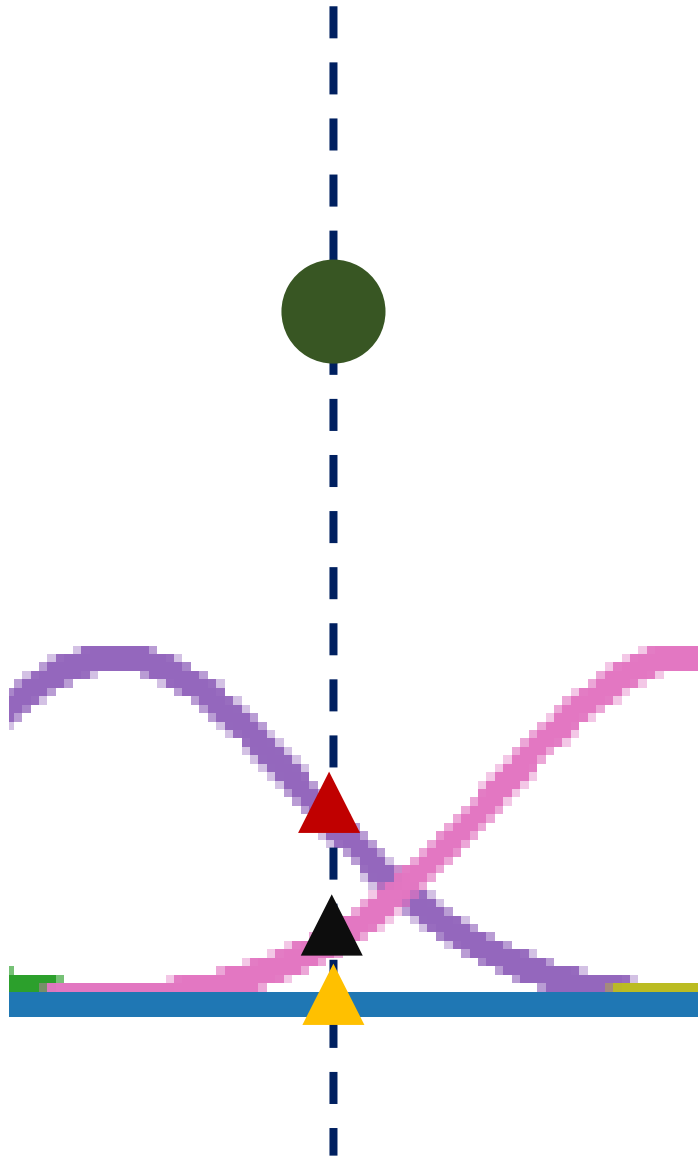








Intersections	Kernel values
Red	$k(3.5)$
Black	$k(5)$
Yellow	$k(2), k(6.5)$



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Red	$k(3.5)$
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$$\begin{aligned}
 Y_{Green} = & y_{x=3.5} \cdot w(3.5) \\
 & + y_{x=5} \cdot w(5) + y_{x=2} \cdot w(2) \\
 & + y_{x=6.5} \cdot w(6.5)
 \end{aligned}$$

Intersections	Kernel values
Red	$k(3.5)$
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$$\begin{aligned}
 Y_{x=Green} = & y_{x=3.5} \cdot w(3.5) \\
 & + y_{x=5} \cdot w(5) + y_{x=2} \cdot w(2) \\
 & + y_{x=6.5} \cdot w(6.5)
 \end{aligned}$$

$$w(x) = \frac{k(x)}{\sum_{i=1}^m k(x_i)}$$

Facts: Kernel regressions

- Non-parametric
 - No math function assumption for the data
 - Flexible
 - Limited usage of mathematics approaches
 - Risk of overfitting

Exercises

Use at least 2 metrics for regression evaluation to compare:

- Linear regression
- Polynomial regression
- Gaussian kernel regression

Don't use regression-support Libraries

$$f(x) = 4 \cos\left(x + \frac{\pi}{10}\right)$$

x	y
1.5	-0.96
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